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Python Programming

From Zero to Programming Hero — Your First Step into Programming

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Task

User input 2 numbers

Arithmetic operation

5

2

Sum 7

Subtraction 3

Multiplication

Division

Reminder

Exponental

Conditional Statements: if, elif, else

Definition: Statements that allow code to make decisions based on conditions

Purpose: Execute different code blocks depending on whether conditions are True or False

Core components: if, elif, else

Syntax uses indentation to define code blocks

```
# Basic syntax
if condition:
    # Code to execute if condition is True
```

```
# Syntax
if condition:
    # Code to execute if condition is True
else:
    # Code to execute if condition is False
```

```
# Syntax
if condition1:
    # Code if condition1 is True
elif condition2:
    # Code if condition1 is False and condition2 is True
else:
    # Code if all conditions are False
```

Common Pitfalls

- Using = (assignment) instead of == (comparison)
- Forgetting the colon at the end of the condition
- Incorrect indentation

```
# Be careful with equality vs assignment
x = 5
if x = 10:  # SyntaxError: use == instead of =
    print("x is 10")

# Forgetting the colon
if x == 5  # SyntaxError: missing colon
    print("x is 5")

# Indentation errors
if x > 0:
print("x is positive")  # IndentationError
```

Practice Exercises

1. Write a program that checks if a number is positive, negative, or zero
2. Create a simple calculator that performs different operations based on user input

Introduction to Loops

- Definition: Loops are used to repeat a block of code multiple times
- Two main types in Python: for loops and while loops
- Purpose: Automate repetitive tasks, iterate through data collections
- Key benefits: Code efficiency, reduced redundancy, data processing

```
# Basic for loop syntax
for variable in sequence:
    # Code to execute for each item
```

```
# Basic while loop syntax
while condition:
    # Code to execute while condition is True
```

Range Function

- The range() function returns a sequence of numbers, starting from 0 by default, and increments by 1 (by default), and stops before a specified number.
- Syntax

  range(start, stop, step)

Parameter	Description
start	Optional. An integer number specifying at which position to start. Default is 0
stop	Required. An integer number specifying at which position to stop (not included).
step	Optional. An integer number specifying the incrementation. Default is 1

The enumerate() Function

- Purpose: Adds a counter to an iterable and returns it as an enumerate object
- Syntax: enumerate(iterable, start=0)
- Returns: A tuple containing count (index) and the value from the iterable
- Why use it? : When you need both the index and value during iteration

```
# Without enumerate
fruits = ["apple", "banana", "cherry"]
index = 0
for fruit in fruits:
    print(f"{index}: {fruit}")
    index += 1

# With enumerate
fruits = ["apple", "banana", "cherry"]
for index, fruit in enumerate(fruits):
    print(f"{index}: {fruit}")

# Output for both:
# 0: apple
# 1: banana
# 2: cherry
```

The enumerate() Function

- The start parameter changes the initial count value
- Useful for human-readable numbering (starting from 1)
- Can start from any integer value

```python
# Starting from 1 instead of 0
fruits = ["apple", "banana", "cherry"]
for index, fruit in enumerate(fruits, start=1):
    print(f"{index}: {fruit}")

# Output:
# 1: apple
# 2: banana
# 3: cherry
```

For `n = 4`, the steps are:

i	Spaces	Stars	Output
1	3	1	`*`
2	2	3	`***`
3	1	5	`*****`
4	0	7	`*******`

Tasks

- The Challenge:Write a program that asks the user to enter a number. Count up from 1 to that number, but apply these two rules using continue and break:
    - If the current number is a multiple of 3, skip printing it using the continue statement.
    - If the current number is greater than or equal to 20, stop the loop entirely using the break statement.

Example Output

If the user enters `25`:

text

```
1
2
4
5
7
8
10
11
13
14
16
17
19
```

Tasks

- The ChallengeWrite a program that loops through days 1 to 7 of a week.
- For each day, use if-elif-else to determine and print the daily schedule based on these rules:
  - On Day 1 (Monday), print: "Gym Day"
  - On Day 3 (Wednesday) or Day 5 (Friday), print: "Study Day"
  - On Day 6 (Saturday) or Day 7 (Sunday), print: "Weekend!
  - Rest Day"For any other day, print: "Work Day"

Expected Output

text

```
Day 1: Gym Day
Day 2: Work Day
Day 3: Study Day
Day 4: Work Day
Day 5: Study Day
Day 6: Weekend! Rest Day
Day 7: Weekend! Rest Day
```

Tasks

- Take a number from user and print it in reverse order using range.
- Print right angled triangle using ‘*’.
- Take a input from user and print inverted right angled triangle

```
*
**
***
****
```

```
Enter the number of rows: 10
**********
*********
********
*******
******
*****
****
***
**
*
```

THANK YOU!

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